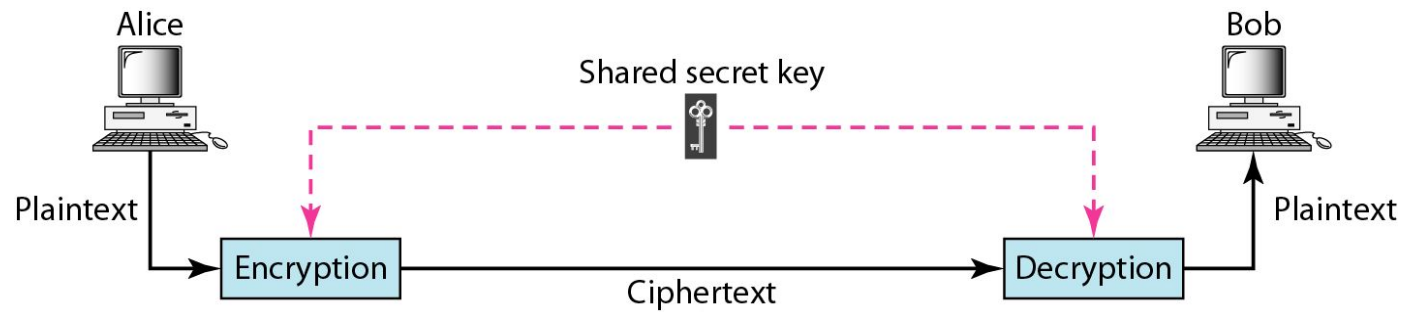
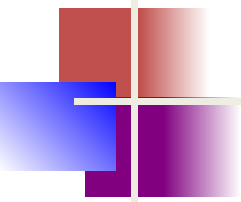


Symmetric and Asymmetric Encryption

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Symmetric-key cryptography





Note

In symmetric-key cryptography, the same key is used by the sender(for encryption) and the receiver (for decryption).

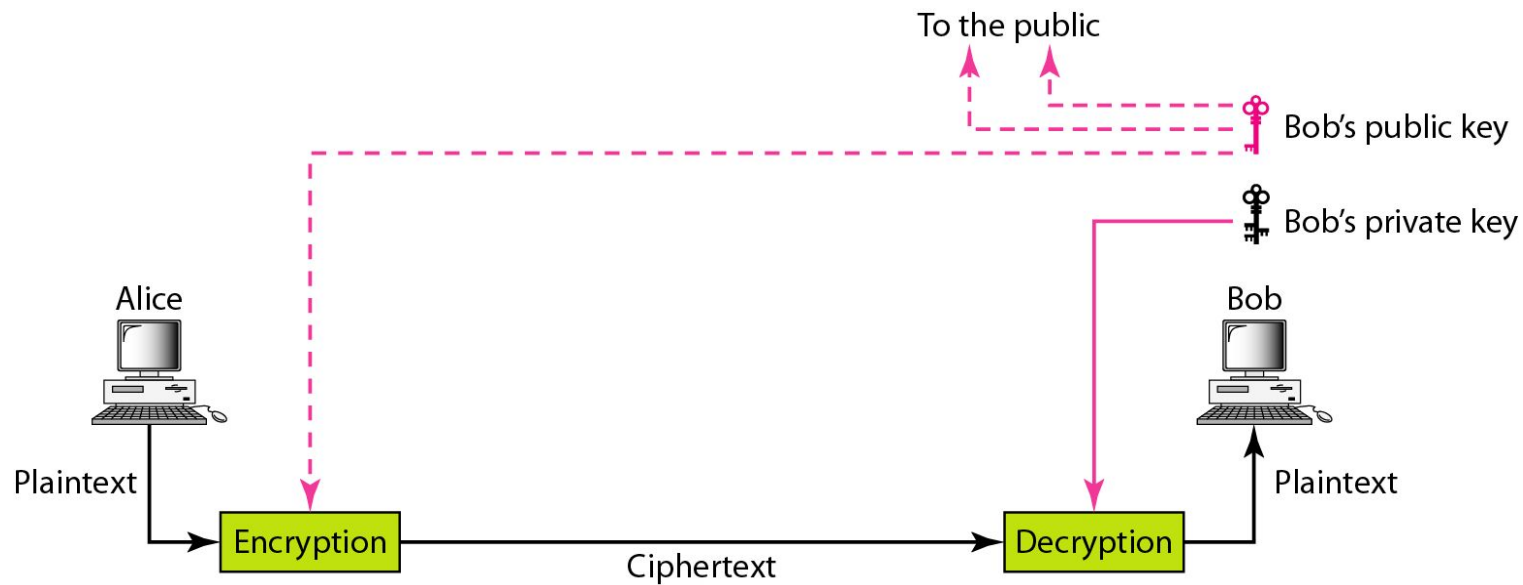
The key is shared.

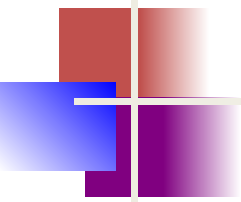
Algorithm: DES,3DES

Symmetric-key cryptography

- Advantages:
 - Simple
 - Faster
- Disadvantages:
 - Key must exchanges in secure way
 - Easy for hacker to get a key as it is passed in unsecure way.

Figure 30.4 *Asymmetric-key cryptography*





Note

An asymmetric-key (or public-key) cipher uses two keys: one private (To encrypt data) and one public (To decrypt data).

- **Asymmetric Key Cryptography (Public Key Cryptography)**
 - 2 different keys are used
 - Users get the Key from an Certificate Authority

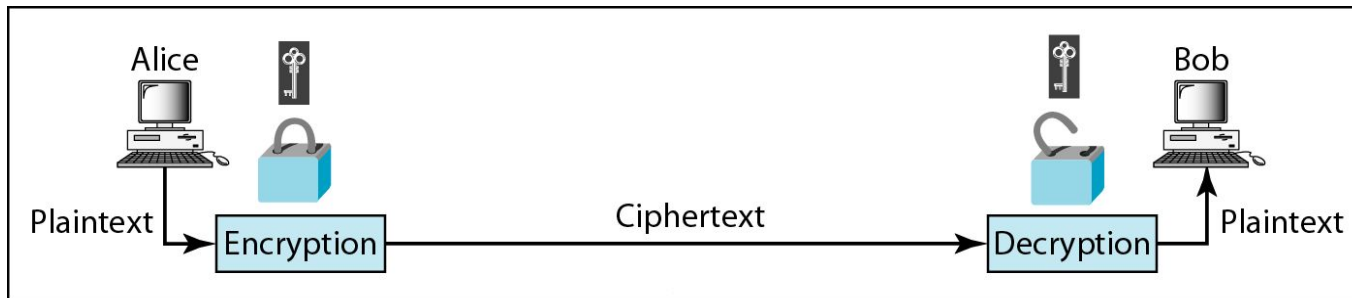
Advantages

- 1. More Secured**
- 2. Authentication**

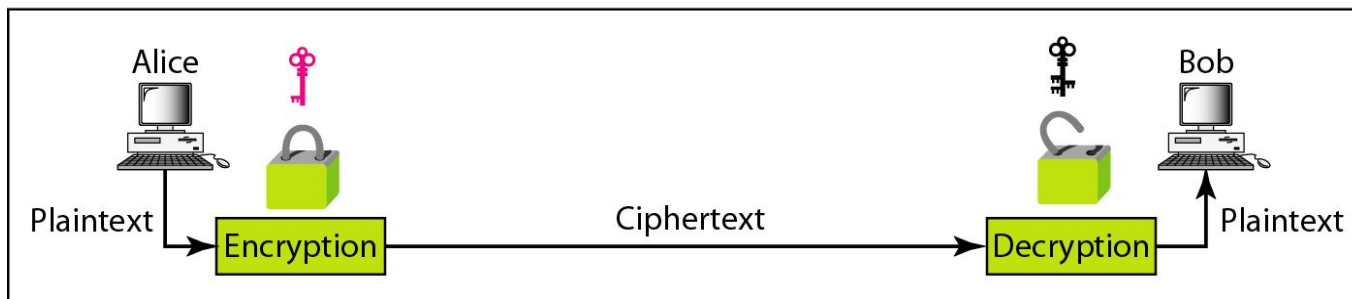
Disadvantages

- 1. Relatively Complex**

Figure 30.6 *Comparison between two categories of cryptography*



a. Symmetric-key cryptography



b. Asymmetric-key cryptography



ASYMMETRIC ENCRYPTION

- Asymmetric encryption use two keys, one to encrypt the data, and another key to decrypt the data.
- These keys are generated together
- One is named as Public key and is distributed freely. The other is named as Private Key and it is kept hidden.
- Both Sender & Recipient has to share their Public Keys for Encryption and has to use their Private Keys for Decryption.

How it WORKS.....?

Step 1: Give your public key to sender.



Step 2: Sender uses your public key to encrypt the plaintext.



plaintext



encryption



ciphertext

Step 3: Sender gives the ciphertext to you.



ciphertext

Step 4: Use your private key (and passphrase) to decrypt the ciphertext.



ciphertext



decryption



plaintext

Key Points in Asymmetric Encryption

- ❖ Asymmetric encryption use two keys:
 - Public Key - to encrypt the data
 - Private Key - to decrypt the data
- ❖ These keys are generated together.
- ❖ The Public key(s) is distributed freely between the sender and receiver.
- ❖ The other is named as Private Key and it is kept hidden.
- ❖ The Private Key is only used for Decryption and will not be shared between the sender and receiver.

Asymmetric Encryption Algorithms

❖ RSA:

❖ Digital Signature Algorithm:

❖ Diffie-Helman:.

MERITS & DE-MERITS

Merits:

- ❖ Two parties **don't need to have** their private keys **already shared** in order to communicate using encryption.
- ❖ **Authentication** and **Non-Repudiation** are **possible**. (Authentication means that you can encrypt the message with my public key and only I can decrypt it with my private key. Non-repudiation means that you can "sign" the message with your private key and I can verify that it came from you with your public key.)

De-Merits:

- ❖ Asymmetric Encryption algorithms are comparatively complex.
- ❖ Time consuming process for Encryption and Decryption.

Categories	Symmetric key Cryptography	Asymmetric key Cryptography
Key used for encryption /decryption	Same key is used for encryption & decryption.	One key is used for encryption & another different key is used for decryption.
Key process	$K_e = K_d$	$K_e \neq K_d$
Speed of encryption/decryption	Very fast	Slower
Size of resulting encrypted text	Usually same as or less than the original clear text size.	More than the original clear text size.
Key agreement/exchange	A big problem	No problem at all.
Usage	Mainly used for encryption and decryption, cannot be used for digital signatures.	Can be used for encryption and decryption as well as for digital signatures.
Efficiency in usage	Symmetric key cryptography is often used for long messages.	Asymmetric key cryptography are more efficient for short messages.

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